Schematic set HM-HW-002: contents

Sheet 1   Index and conventions

Sheet 2   Analog front end (custom part)

Sheet 3   ESP32 and peripherals

Sheet 4   Power and CHARGE / USE isolation

Conventions

- Signals flow left to right. Net flags (arrow boxes) are global labels: the same name on any sheet is one net.
- Power nets: +3V3 (blue), +5V (orange), VBAT 3.0-4.2 V (red), VREF 1.65 V (violet), GND (grey).
- Values match the KiCad project hardware/kicad/ (ERC 0, netlist path check) and the LTspice netlists in sim/circuit/.
- Patient side = applied part, IEC 60601-1 type BF. Battery only while a person touches the electrodes (SW6 on USE).
- Never on anyone with a pacemaker or ICD (weight-only mode). Monitoring, not diagnosis.
- Related documents: HM-HW-001 block diagram · HM-HW-003 power budget · HM-HW-004 PCB · HM-HW-005 BOM · HM-HW-006 verification.

Revision history

A	2026-10-25	first release: 8-electrode AFE, 3-frequency spectroscopy, Rcal self-calibration, CHARGE/USE isolation

HridayMitra stool: Index and conventions

Doc HM-HW-002   Rev A   2026-10-25   Size A3

Sheet 1 / 4   Checked: KiCad ERC 0 · LTspice · firmware emulator

DAC_EXC ← ESP32 GPIO25 (sheet 3)

Z_ADC → ESP32 GPIO35 (sheet 3)

PATH_S0, PATH_S1, SEL, CAL_SEL ← ESP32 (sheet 3)

1 Current source (100 µA)

2 Path switch + body (legs path)

3 Sense pair + 10 MΩ bias

4 Select + INA126 + anti-alias

ESP32 DAC GPIO25 5 / 20 / 50 kHz

TP1

C2 1 µF

U1B MCP6002

R3 100 k

VREF

TP2

C3 1 µF DC block

R4 4.7 k 1 % current limit

TP3

U4 74HC4052 current pair

X

X0

Y

Y0

S0 GPIO33, S1 GPIO27

X1-3 / Y1-3: arms, sides

LF_FORE plate

Rc 1 k skin

Rc 1 k (LF_HEEL)

left leg Re ‖ (Ri+Cm)

pelvis

right leg Re ‖ (Ri+Cm)

Rc 1 k (RF_HEEL)

Rc 1 k skin

RF_FORE plate

U5 74HC4052 sense pair

X0

X

Y0

Y

R6 1 k

SA

TP4 = SA - SB

R8 10 M

VREF

SB

R7 1 k

R9 10 M

VREF

U3 74HC4053 SEL GPIO18

A0

A

A1

B1

B0

B

RTAP

U2 INA126 G = 10 RG 16 k

TP5 / TP7

TP6

ESP32 ADC GPIO35

R11 1.5 k

C5 10 n

ESP32: DMA 200 kS/s, Goertzel at 5 / 20 / 50 kHz

Z = Rcal × |TP5| / |TP7| (amplifier gain and current cancel)

Reference chain (self-calibrating)

RET

NCAL

VREF

C4 1 µF DC block

R5 220

Rcal 100 Ω 0.1 %

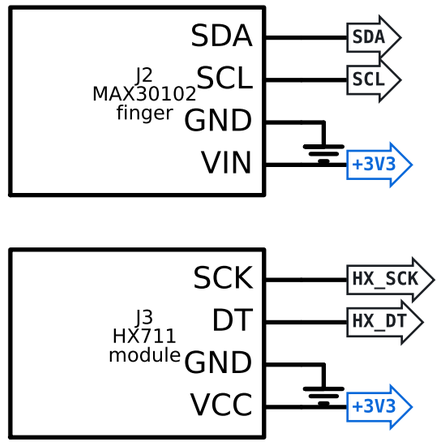
RTAP = NCAL or RET (U3 switch C, GPIO12): the reference across Rcal, or across R5 + Rcal for the self-check (ratio 2.2)

Patient safety (IEC 60601-1 type BF, LTspice): C3 and C4 in series block DC; R4 limits the current; every single fault stays ≥ 12.8× under the limit. CHARGE / USE sw … tead of

HridayMitra stool: Analog front end (custom part)

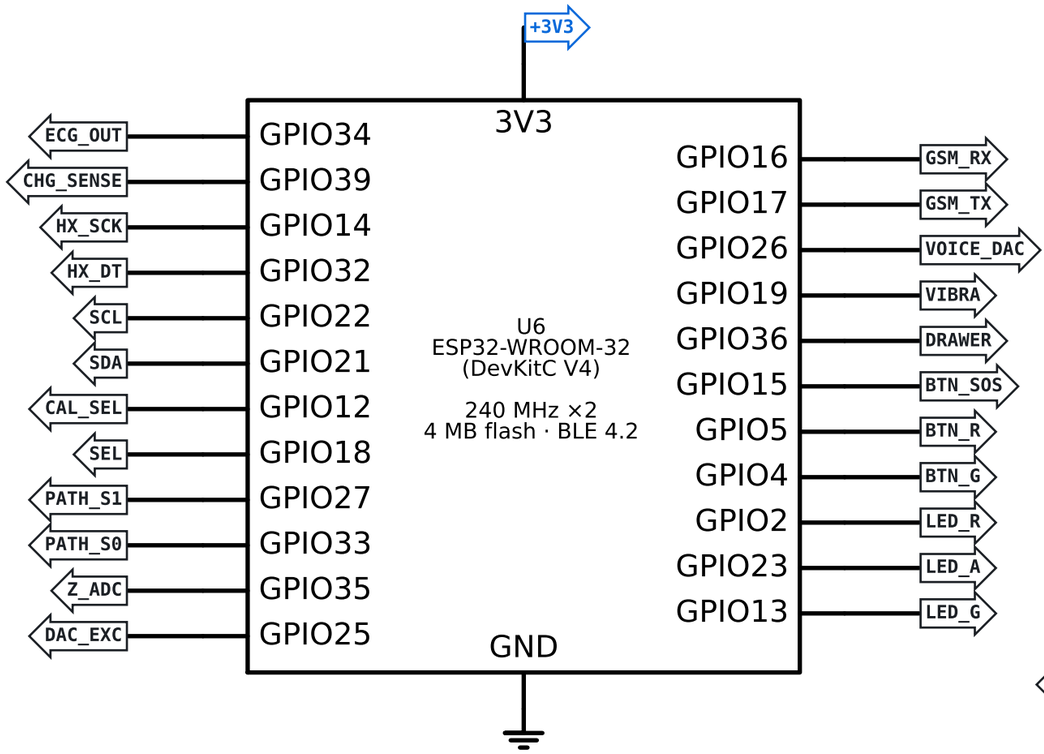
Doc HM-HW-002 Rev A 2026-10-25 Size A3

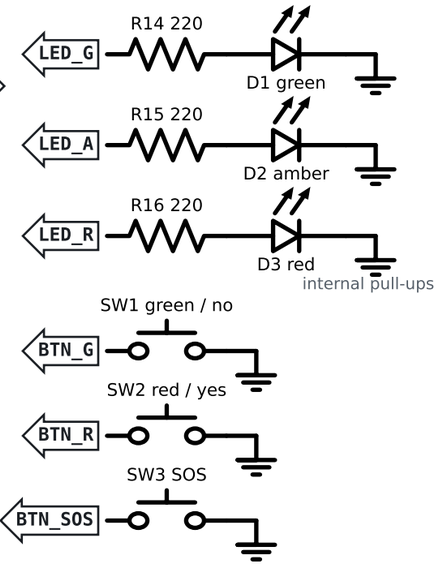
Sheet 2 / 4 Checked: KiCad ERC 0 · LTspice · firmware emulator



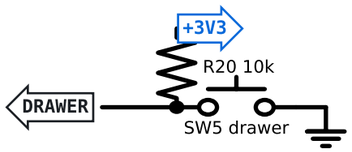
J4/J5: HX711 E+ E− A− A+ ← 4 × 50 kg half-bridge load cells, wired as one full bridge

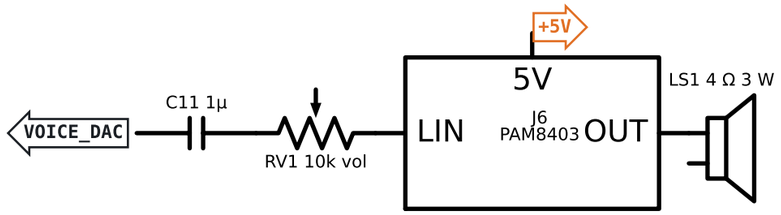
To sheet 2 (AFE): DAC_EXC, Z_ADC, PATH_S0, PATH_S1, SEL, CAL_SEL. GPIO12 is a boot strap pin: only the 74HC4053 select input on it, no pull-up. ECG_OUT: optional AD8232, not fitted.

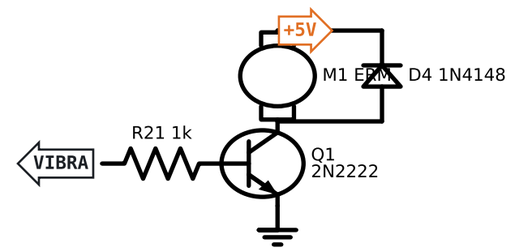


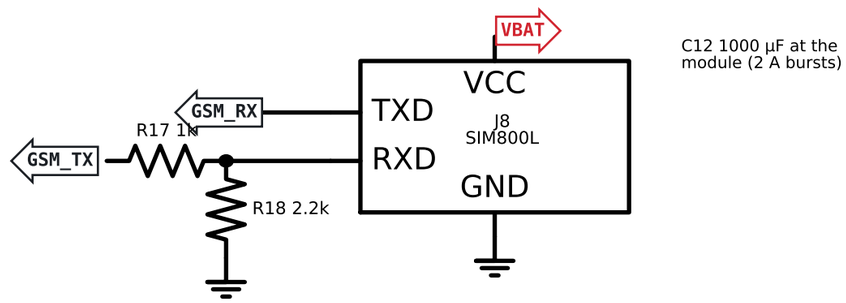


On the DevKit: EN pull-up R19 10k · CP2102 USB-UART (laptop / app) · AMS1117 (sheet 4)









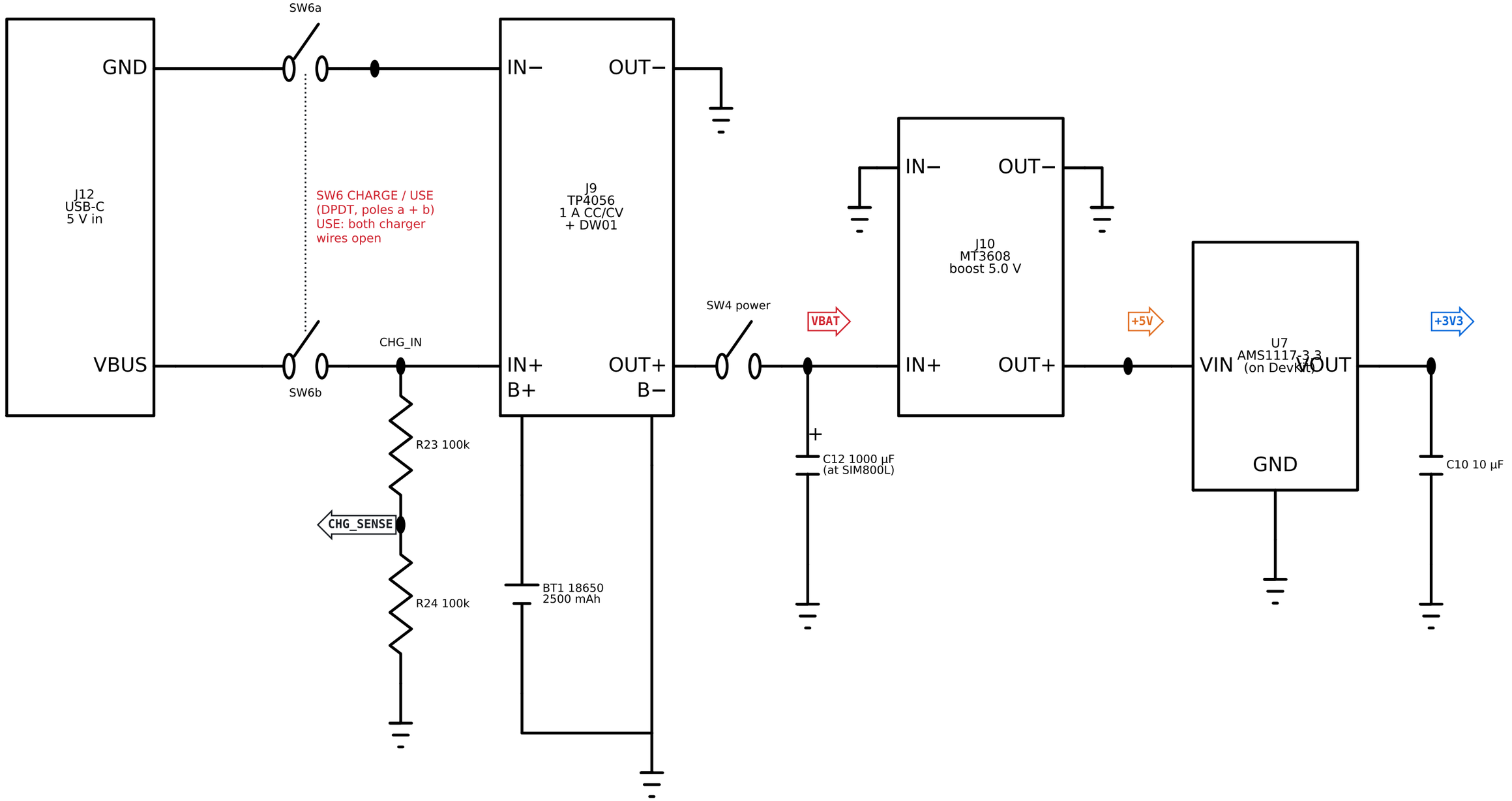
C12 1000 µF at the module (2 A bursts)

R17/R18: 3.3 V → 2.3 V into the SIM800L's 2.8 V RXD

HridayMitra stool: ESP32 and peripherals

Doc HM-HW-002 Rev A 2026-10-25 Size A3

Sheet 3 / 4 Checked: KiCad ERC 0 · LTspice · firmware emulator



USE (normal): both charger wires open; the stool runs on BT1 only. LTspice leakage through an earthed patient: 0.19 µA (339 µA with a cheap charger connected, 72 mA if its insulation fails). CHARGE: CHG_IN high → R23/R24 → GPIO39 → the firmware refuses to measure. The TP4056 module's own USB socket is sealed: J12 + SW6 is the only way in.

Rails: VBAT → SIM800L directly (2 A bursts, C12 at the module) · +5V → DevKit VIN, PAM8403, ERM motor · +3V3 → AFE, MAX30102, HX711, LEDs · VREF 1.65 V is made on sheet 2 (R1/R2 + U1A).
Power budget HM-HW-003: 130 mA idle as built (0.7 days per charge); Rev B plan 22 mA (3.9 days).

HridayMitra stool: Power and CHARGE / USE isolation

Doc HM-HW-002 Rev A 2026-10-25 Size A3

Sheet 4 / 4 Checked: KiCad ERC 0 · LTspice · firmware emulator